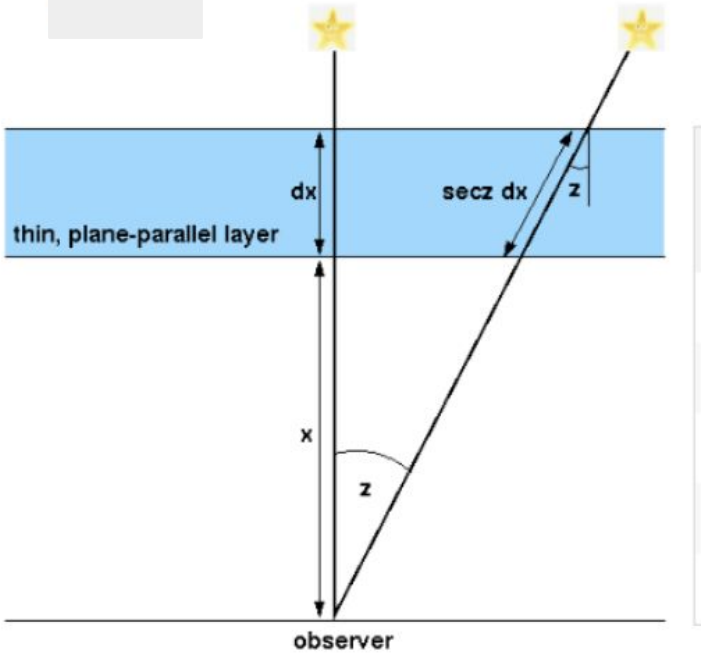
Zero Point Calibration

- The **instrument magnitude** (in arbitrary units) is given by: $$m_{inst} = -2.5 \log\left(\frac{flux}{t_{\exp}}\right)$$

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- **Atmospheric extinction correction**: if target and standard stars are not in the same field, i.e. have different airmass – atmospheric extinction is not the same:



$$m_{inst} = m_{inst,0} + k \sec z$$

Filter	*k*
B	0.25
V	0.15
R	0.09

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$$m_{inst} = m_{inst,0} + k \sec z$$

- **Standard star**: compare to literature value: $m_{ZP} = m_{std} - m_{std,0}$

- **Calibrated magnitude of targets:** $m_{calib} = m_{inst,0} + m_{ZP}$